USER MANUAL & TECHNICAL GUIDE

ROBOCal 9-Sensor Array

9-Channel Infrared Reflective Sensor Array for Robotics & Line Following



Document detail	Value
Model number	**MS-R9IR-V1**
Product version / revision	**[PRODUCT VERSION / REVISION: _____]**
Manual version	**[MANUAL VERSION: _____]**
Date	**[DATE: _____]**
Website / contact / social media	**[WEBSITE / CONTACT / SOCIAL MEDIA: _____]**

About This Document

Three kinds of information appear in this manual. They are kept visibly separate so readers always know how much to trust a number.

Type	Meaning	How it appears
Verified specification	Confirmed on the final PCB revision or by measurement.	Plain text in tables
Recommended value	Guidance from the designer, based on design intent.	Text marked "recommended" or "approx."
Experimental / TBD	Not yet measured or confirmed.	**[Yellow highlighted placeholder]**

> **NOTE** Figures are numbered to match the list in Section 27. Where an image is not yet available a dashed placeholder box is shown instead. No photographs or app screenshots have been fabricated.

1. Product Introduction

The **9-Sensor Array** is a 9-channel infrared (IR) reflective sensor board built for robotics, and especially for line-following robots. It was designed to give a wider sensing area and better coverage than the common 5-sensor TCRT5000-style modules that most beginners start with.

Why a 9-sensor array?

Many inexpensive line-following boards available to beginners carry only about five sensors. Larger arrays do exist, but some use very small SMD infrared components that have very short sensing distances. That makes mounting and ground clearance harder on some robots. The ROBOCal array instead uses **nine individually positioned IR emitter/receiver pairs** spread across a relatively wide sensing area, using approximately 5 mm components.

Intended applications

- Line-following robots
- High-speed line followers
- Autonomous robots
- Maze-solving robots
- Edge / track detection
- Reflective surface detection
- Robotics experiments and educational robotics
- Custom ESP32-based robots

> **IMPORTANT** Actual performance depends on the surface, line colour, ambient lighting, sensor height, robot speed, calibration, electronics and software. This manual does not promise a specific result under any condition.

Recommended system

The recommended architecture is **9-Sensor Array → ESP32-S3 → Wi-Fi → ROBOCal App**. The ESP32-S3 can read all nine sensors on its ADC1 channels while Wi-Fi stays active. Classic ESP32 boards and the Arduino Nano (with an external multiplexer) are supported as compatibility options, described in Sections 12 to 14.



Figure 1 – 9-Sensor Array product overview

2. Product Design Philosophy

The board carries **9 infrared sensing channels**. Each channel consists of one infrared emitter and one infrared receiver. The layout is arranged to give a broad sensing footprint while keeping useful spacing between neighbouring channels.

Design parameter	Value	Status
Sensing channels	9	Design
Emitters / receivers per channel	1 + 1	Design
Sensor-to-sensor centre spacing	Approx. 15 mm	Approximate
Expected optical / package area	Approx. 8–12 mm	Approximate
Emitter / receiver size	Approx. 5 mm each	Design intent

The 8–12 mm optical area follows from the intended use of approximately 5 mm IR transmitter and 5 mm IR receiver components. Final dimensions depend on the exact components chosen for production.

[INSERT FINAL VERIFIED SENSOR-TO-SENSOR CENTER SPACING: _____ mm]

NOTE The combination of roughly 15 mm spacing and a PCB about 120 mm long is the main practical difference from typical 5-sensor modules: it covers a wider strip of track at once.

3. Physical Dimensions

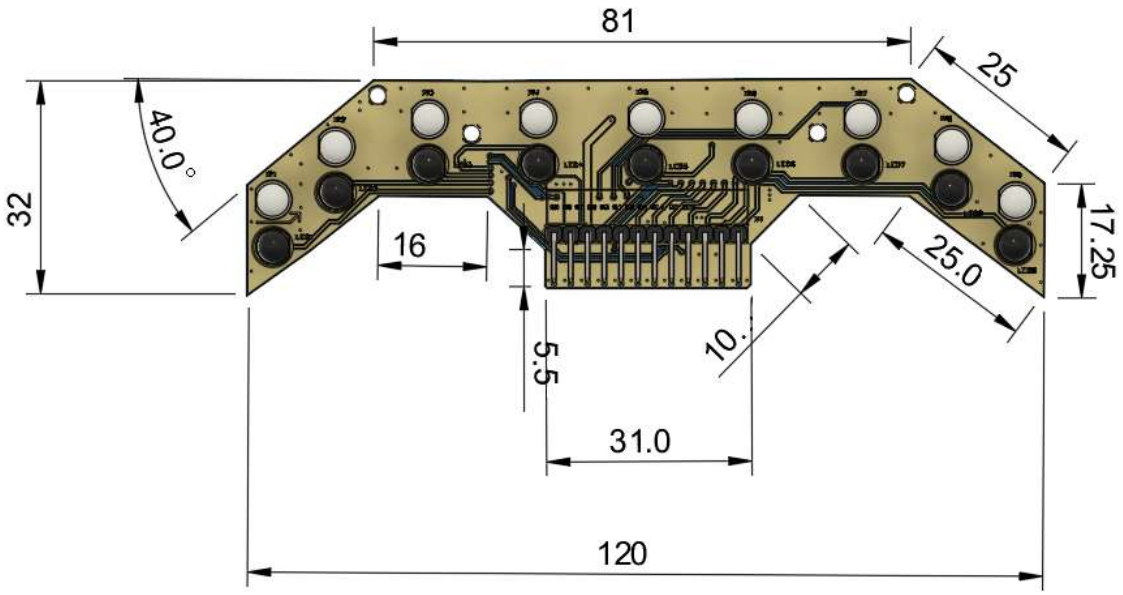


Figure 2 – ROBOCal 9-Sensor Array Mechanical Dimensions

The supplied engineering drawing shows the following approximate dimensions:

Feature	Approximate dimension
Overall PCB length	Approx. 120 mm
Main sensing section	Approx. 81 mm
PCB height at main body	Approx. 32 mm
Side / end geometry	Approx. 25 mm
Central connector region	Approx. 32 mm
Connector / sensor-area dimensions	As shown in the engineering drawing

IMPORTANT Drawing annotations are not manufacturing tolerances unless explicitly verified. All dimensions should be verified against the final PCB revision before mechanical integration.

4. Sensor Arrangement

On the front of the board you will see two kinds of component. The **white (clear) components are the IR emitters (transmitters)**, and the **black components are the IR receivers**. Each sensing channel pairs one emitter with one receiver.

How reflective IR sensing works

Each emitter sends infrared light down toward the surface under the robot. The matching receiver measures how much of that light bounces back. Surfaces reflect differently, so the reading tells the robot what is underneath:

- **Light surface** → generally stronger reflection
- **Dark line** → generally weaker reflection
- The electronics turn this difference into an **analog signal** that your controller reads

> **IMPORTANT** The output polarity (whether white reads high or low) depends on the receiver circuit and your software. Do not assume a fixed ADC value for black or white. Always calibrate the actual board (Section 15).

5. Ambient Light / Optical Interference Design

Reflective IR sensors can be disturbed by external infrared or ambient light, because the receiver cannot tell the difference between light that came from its own emitter and light from the room or the sun.

On this board, the emitters and receivers are positioned with the intention of reducing the effect of external light entering directly from the front. The physical arrangement tries to make it less likely that unwanted incoming light produces a dominant signal that has nothing to do with the surface reflection.

> **WARNING** "The optical arrangement is intended to reduce the influence of external light, but it does not provide complete immunity to sunlight or other strong infrared sources."

Always calibrate under the actual operating environment, with the same lighting you will use when the robot runs.

6. IR Emitter Control – L Pin

The **L pin** controls the infrared emitter supply/control circuit through an **AO3400A N-channel MOSFET**. The AO3400A acts as the switching element for the emitter section.

This allows your controller to:

- Turn the IR emitters ON and OFF
- Control the emitter drive condition
- Potentially implement software-controlled illumination strategies
- Reduce unnecessary emitter operation

> **NOTE** The L pin is described as an ON/OFF switching control. This manual does not claim true analog brightness control; that claim should only be added if the final circuit is verified to provide it.

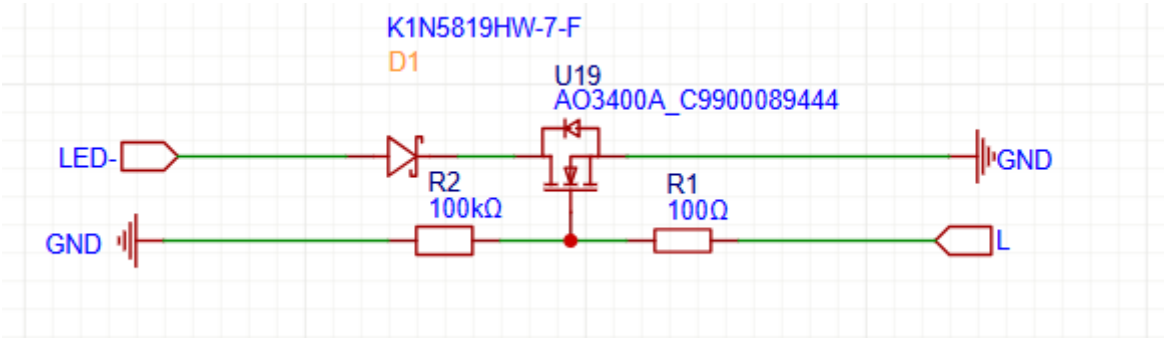


Figure 8 – L-pin circuit

7. Emitter Bypass / Always-ON Option

The PCB includes a small solderable bypass pad (jumper) option for users who do not want software-controlled emitters and prefer to keep the emitter section permanently enabled.

Configuration	Behaviour
Normal (default)	The controller switches the emitters through the L pin.
Bypass	Solder the designated pads to force emitter control into the always-enabled condition.

IMPORTANT The exact electrical state of the bypass pads must be described according to the verified schematic. It is intentionally not stated here.



Figure 9 – Emitter Control Bypass Pads

WARNING "Only bridge the designated bypass pads. Do not short unrelated pads or traces."

8. Recommended Sensor Height

Mounting height is one of the most important performance parameters of this array. The optimum height must be found by experiment, using the final emitter and receiver components, the final PCB, your surface material, the line width, the ambient lighting and your robot speed.

Recommended Sensor-to-Surface Height

Parameter	Value
Final tested optimum height	**[FINAL TESTED OPTIMUM HEIGHT: _____ mm]**
Tested operating range	**[TESTED OPERATING RANGE: _____ to _____ mm]**
Recommended maximum height	**[RECOMMENDED MAXIMUM HEIGHT: _____ mm]**

In general, changing the distance changes the reflected signal level, and a greater distance may widen the sensing footprint. A sensor mounted too close can also behave badly, depending on the optical characteristics of the components. Perform calibration after physically mounting the array.

> **WARNING** No sensing distance is promised until it has been experimentally validated.

[INSERT IMAGE HERE – FIGURE 10: RECOMMENDED MOUNTING HEIGHT]

Figure 10 – Recommended mounting height

9. Electrical Interface

The array provides nine sensor outputs, **S1 to S9**, along with **VCC**, **GND** and the **L** (emitter-control) input. The connector in the supplied image is labelled approximately:

`GND VCC L S1 S2 S3 S4 S5 S6 S7 S8 S9`

Pin	Function	Description	Controller connection
S1	Sensor 1	Analog sensor output	_____
S2	Sensor 2	Analog sensor output	_____
S3	Sensor 3	Analog sensor output	_____
S4	Sensor 4	Analog sensor output	_____
S5	Sensor 5	Analog sensor output	_____
S6	Sensor 6	Analog sensor output	_____
S7	Sensor 7	Analog sensor output	_____
S8	Sensor 8	Analog sensor output	_____
S9	Sensor 9	Analog sensor output	_____
L	Emitter control	IR emitter control	_____
GND	Ground	Common ground	_____

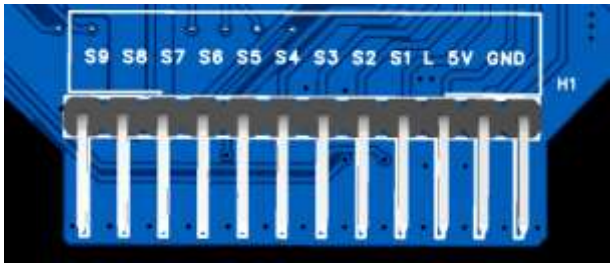


Figure 4 – Connector pinout

Electrical parameter	Value
Operating voltage	**5V**
Maximum current	**[INSERT MAXIMUM CURRENT: _____ mA]**
Sensor output voltage range	**[INSERT SENSOR OUTPUT VOLTAGE RANGE: _____ V]**

NOTE These values must be filled from the final tested circuit, not guessed. Note that a VCC pin is part of the interface but is not shown in the approximate label sequence above; confirm its position on the final PCB.

10. Recommended Controller – ESP32-S3

The **ESP32-S3 is the primary recommended controller.** It provides enough ADC1 channels to read all nine sensor outputs directly while Wi-Fi remains available.

Signal	Connect to
9 sensor outputs (S1–S9)	ESP32-S3 ADC1 inputs
L	An ESP32-S3 GPIO
GND	ESP32-S3 GND
VCC	Appropriate supply (per verified electrical specification)

Because all nine inputs sit on ADC1, the ESP32-S3 can run the following at the same time:

- The 9-sensor array
- Wi-Fi
- The companion ROBOCal Android application
- Calibration functions
- Live sensor visualization
- PID tuning
- Robot control

Item	Value
Recommended ESP32-S3 board / module	**[INSERT RECOMMENDED ESP32-S3 BOARD/MODULE MODEL: _____]**
ADC1 GPIO mapping	**[INSERT EXACT ADC1 GPIO MAPPING: _____]**
L control GPIO	**[INSERT L CONTROL GPIO: _____]**

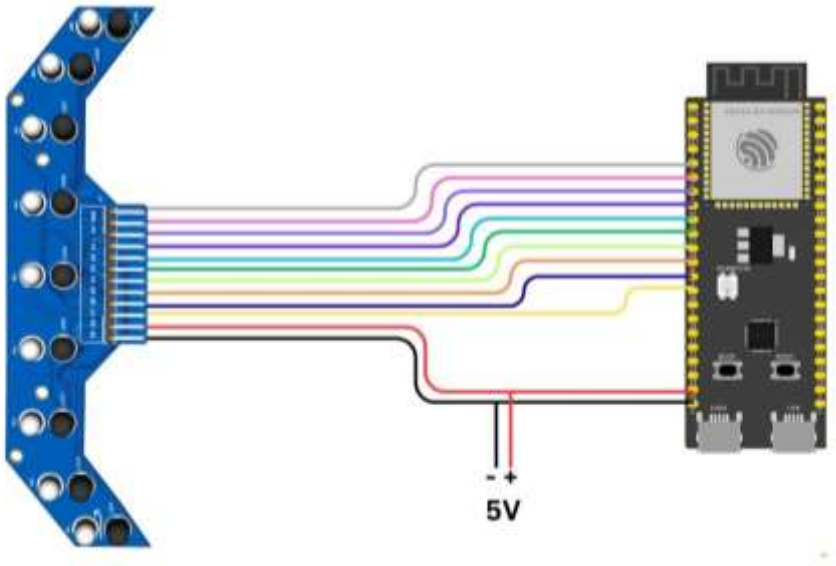


Figure 5 – ESP32-S3 connection

11. ESP32-S3 + ROBOCal App

The array is intended to work with the accompanying ESP32-based robotics application. Together they can provide:

- Live sensor values
- Sensor calibration
- Sensor visualization
- Threshold adjustment
- PID parameter adjustment
- Line-following testing
- Robot control
- Wireless configuration

[INSERT APP SCREENSHOT – SENSOR CALIBRATION]

[INSERT APP SCREENSHOT – LIVE SENSOR VALUES]

[INSERT APP SCREENSHOT – PID CONTROL]

Figure 12 – App interface

[INSERT APP DOWNLOAD / QR CODE]

12. Classic ESP32 DevKit / ESP32-WROOM Compatibility

The array can also be used with classic ESP32 development boards and ESP32-WROOM-based boards, but there is an important ADC limitation to understand first.

The classic ESP32 offers fewer conveniently usable ADC1 channels for this application, and its ADC2 has limitations while Wi-Fi is active.

> **WARNING** If all 9 sensors are connected directly to the classic ESP32 ADC inputs, Wi-Fi operation may not be available simultaneously, depending on the selected ADC channels and implementation.

You therefore choose between two options:

Option	Approach	Trade-off
A – Maximum direct sensor inputs	Use the available ADC channels directly.	No Wi-Fi-dependent operation.
B – Wi-Fi + all 9 sensors	Use an external analog multiplexer / ADC or another input-expansion solution.	Extra part and scanning software.

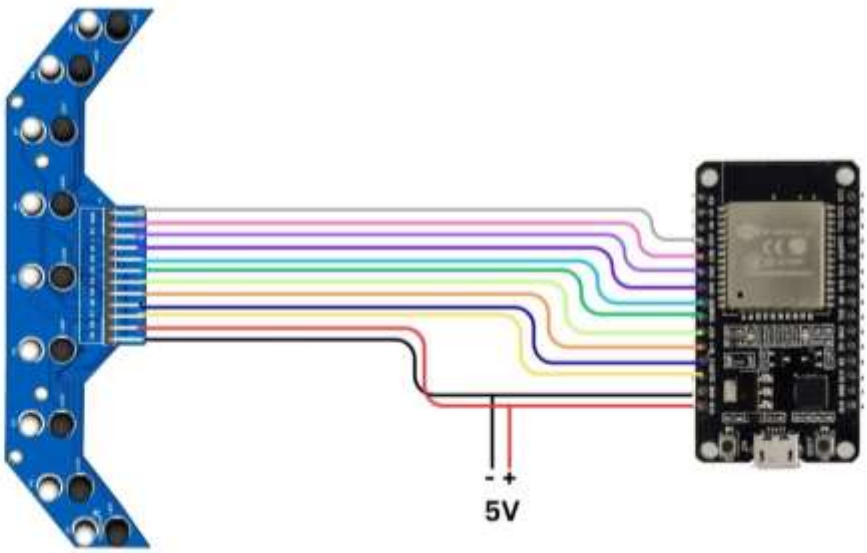


Figure 6 – ESP32-WROOM / DevKit connection without Wi-Fi

13. Arduino Nano Compatibility

The ROBOCal array can also be used with an **Arduino Nano**. The Nano does not have nine independent analog inputs, so an **external analog multiplexer is required** to read all nine sensors.

9 sensor outputs → Analog multiplexer → Arduino Nano analog input

The multiplexer lets the Nano select one sensor at a time and read its analog value. This makes the array usable with a much wider range of robots and controllers.

Because the sensors are scanned one after another instead of being sampled at the same instant, the Nano implementation may need extra software handling.

Item	Value
Recommended multiplexer part number	CD74HC4067
Multiplexer pin mapping	**[INSERT MULTIPLEXER PIN MAPPING: _____]**

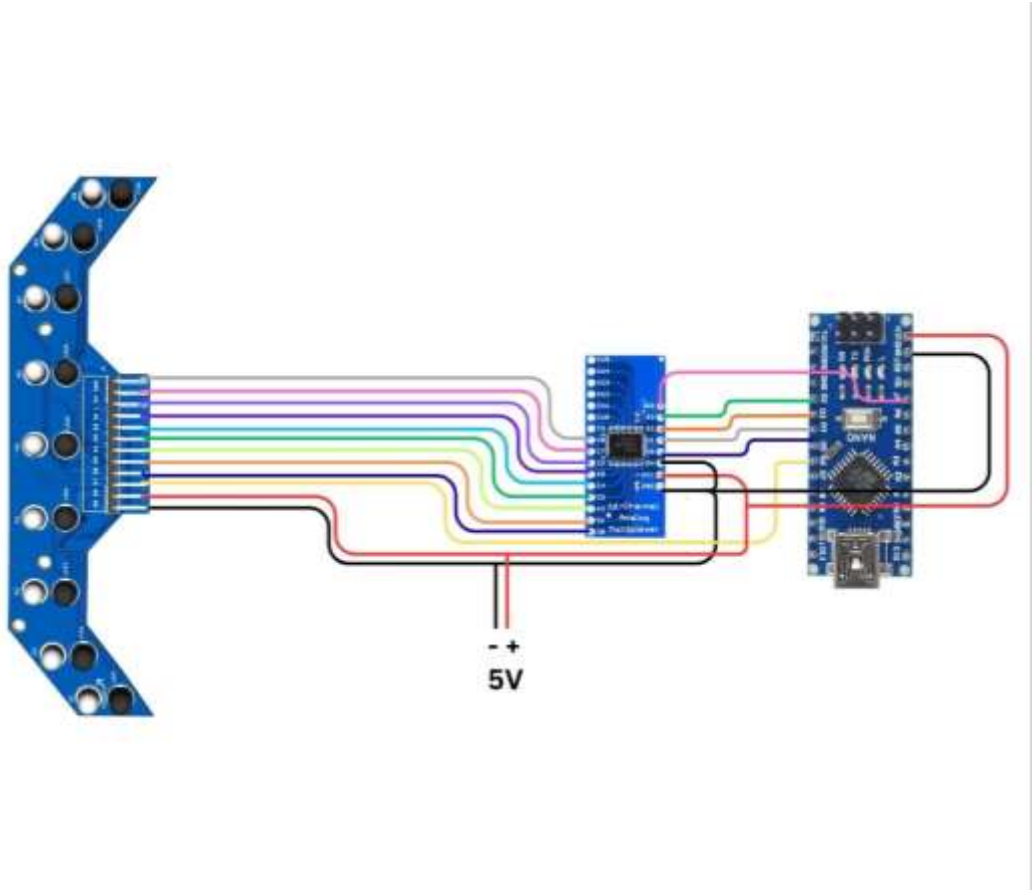


Figure 7 – Arduino Nano + multiplexer connection

14. Arduino Compatibility Philosophy

The PCB intentionally does **not** include an onboard multiplexer. Reasons:

- Keeps the sensor PCB simple
- Reduces component count and cost
- Makes the board easier to understand
- Keeps the sensor array controller-independent
- Lets users choose their own multiplexer or ADC solution
- Makes it directly convenient for ESP32-S3 users

A separate application note will show how to connect the array to an Arduino through a multiplexer.

15. Calibration Procedure

Calibration teaches your software what "white" and "black" look like for your exact board, height, surface and lighting. Follow these steps in order.

1. Mount the array at the recommended height.
2. Connect VCC and GND.
3. Connect the nine sensor outputs.
4. Turn on the IR emitters (via the L pin, or by using the bypass option).
5. Place the array over the white / background surface.
6. Record the sensor values.
7. Place the array over the line / dark surface.
8. Record the sensor values.
9. Calculate thresholds or normalized values.
10. Test the complete line width.

Example value	Entry
White surface typical value	**[White surface typical value: _____]**
Black line typical value	**[Black line typical value: _____]**
Recommended threshold	**[Recommended threshold: _____]**
Recommended calibration method	**[Recommended calibration method: _____]**

IMPORTANT Calibrate under the same lighting and mounting conditions that the robot will use in actual operation. Recalibrate if the surface, lighting or height changes.

[INSERT IMAGE HERE – FIGURE 11: CALIBRATION EXAMPLE]

Figure 11 – Calibration example

16. Sensor Numbering and Physical Orientation

Sensors are numbered **S1** → **S9** across the array. Getting the orientation right prevents your software from reading the line position mirrored Right-to-Left.

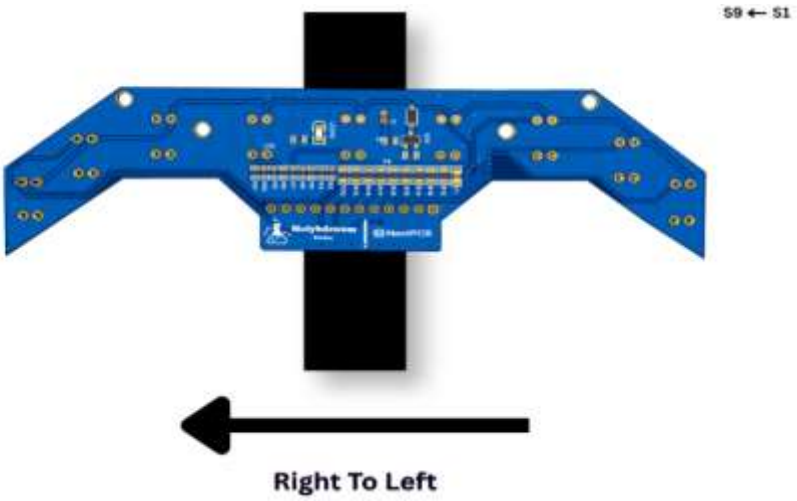


Figure 3 – Sensor numbering

> **NOTE** Define clearly which end of the array is the robot's left side before writing position-calculation code.

17. Recommended Robot Integration

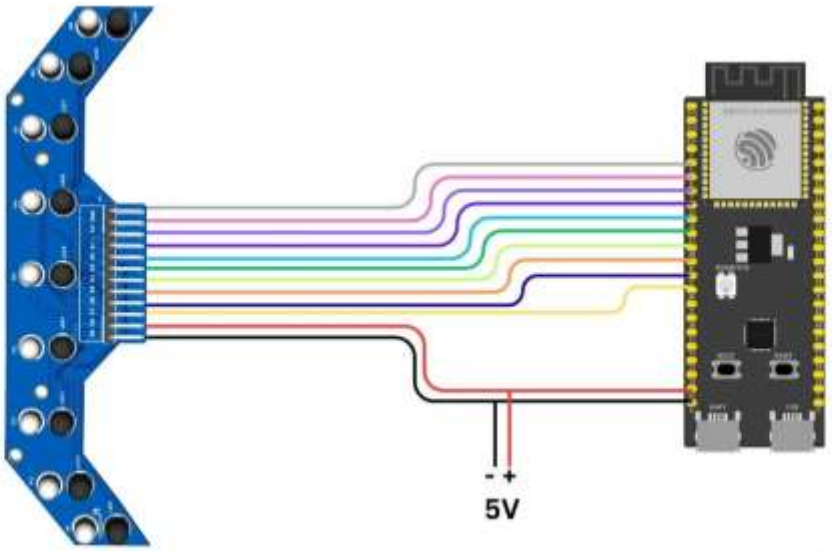
Good mechanical installation matters as much as good code. Consider the following:

- **Mounting bracket design** – rigid, repeatable, easy to adjust in height
- **Sensor-to-ground clearance** – set and re-check at the tested height
- **PCB protection** – keep the board from scraping or catching on the track
- **Cable routing** – keep wires clear of wheels and moving parts
- **Connector strain relief** – secure cables so they do not pull on the connector
- **Vibration** – avoid mounting that lets the board flex or rattle
- **Alignment** – keep the board parallel to the ground
- **External IR** – avoid direct exposure to strong external IR sources

Parameter	Value
Recommended mounting height	**[INSERT RECOMMENDED MOUNTING HEIGHT: _____ mm]**
Recommended line width	**[INSERT RECOMMENDED LINE WIDTH: _____ mm]**
Recommended robot speed range	**[INSERT RECOMMENDED ROBOT SPEED RANGE: _____]**

18. Example Connection – ESP32-S3

ROBOCal	ESP32-S3
S1	GPIO 4
S2	GPIO 5
S3	GPIO 6
S4	GPIO 7
S5	GPIO 15
S6	GPIO 16
S7	GPIO 17
S8	GPIO 18
S9	GPIO 8
L	GPIO 46
VCC	5v
GND	GND



19. L-Pin Control Examples

Software-controlled emitters

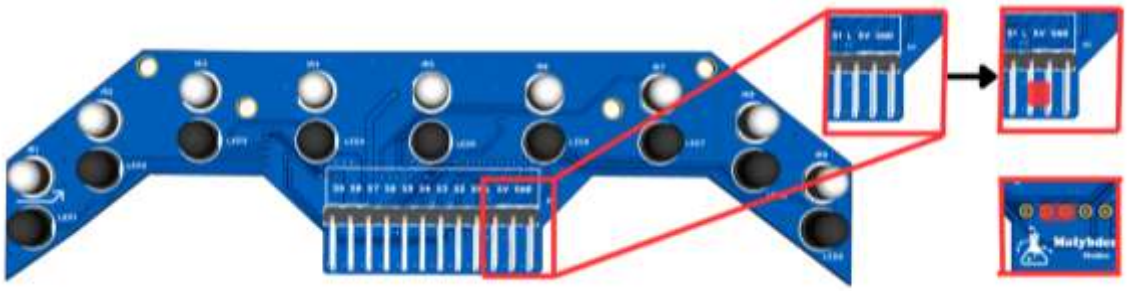
Connect a controller GPIO to **L**. The software decides when the emitters are on.

Always-ON configuration

Short the designated pins (see Section 7).

Testing configuration

Use the controller to toggle the emitter section during calibration and debugging, for example to compare readings with the emitters on and off and see how much ambient light contributes.



Gate input	P-channel MOSFET	LED state
HIGH	OFF	OFF
LOW	ON	ON

20. Troubleshooting

Problem	Possible cause	Solution
All sensors read incorrectly	Power / GND problem	Check supply and ground
One sensor behaves differently	Component / PCB issue	Inspect sensor and soldering
Values change in sunlight	Ambient IR interference	Shield, calibrate or change mounting
Sensor values saturate	Sensor too close / strong reflection	Adjust height
Very weak response	Sensor too far / low emitter drive	Check height and emitter circuit
L pin does not work	GPIO / wiring / configuration	Check MOSFET control
Wi-Fi stops working on classic ESP32	ADC2 / Wi-Fi conflict	Use ADC1 or an external multiplexer / ADC
Arduino cannot read all sensors	Insufficient analog inputs	Use an analog multiplexer

21. Electrical and Safety Notes

WARNING Read before connecting power.

- Do not exceed the specified supply voltage (5v).
- Confirm polarity before connecting power.
- Do not short VCC and GND.
- Only bridge the designated bypass pads.
- Do not assume every 5 mm IR emitter or receiver has identical electrical characteristics.
- Verify component specifications before replacing sensors.
- Disconnect power before modifying solder jumpers.

22. Technical Specifications

Parameter	Specification
Model number	MS-R9IR-V1
Number of sensors	9
Sensor type	IR reflective
Emitter size	Approx. 5 mm
Receiver size	Approx. 5 mm
Sensor spacing	15 mm (approximate; verify)
Expected sensor optical / package area	8–12 mm
PCB length	120 mm (approximate; verify)
PCB height	Approx. 33 mm
Sensor outputs	9 analog outputs
Emitter control	L pin

Parameter	Specification
Emitter switching device	AO3400A
Recommended controller	ESP32-S3
Other controller	ESP32 / ESP32-WROOM
Arduino support	Arduino Nano + multiplexer
Operating voltage	5v
Current consumption	[TBD]
Recommended sensor height	[TBD]
Operating temperature	-10° to 80° C

23. Design Limitations

- It is an IR reflective sensor, not a camera.
- Strong sunlight can affect readings.
- Sensor height affects performance.
- Different floor materials produce different readings.
- Calibration is recommended.
- The array does not guarantee perfect line detection at every height or speed.
- Classic ESP32 ADC / Wi-Fi limitations must be considered.
- The Arduino Nano requires external input expansion for all nine sensors.

24. Future Expansion

> **NOTE** The items below are ideas for possible future versions. They are not features of the current product.

- Integrated multiplexer version
- Digital comparator outputs
- Adjustable emitter current
- Improved optical shielding
- Smaller / lighter version
- Higher-speed sampling
- Integrated calibration memory
- Different sensor spacing options